

CS229 Problem Set 0 Solutions

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1. Gradients and Hessians

Given $f(x) = \frac{1}{2}x^T Ax + b^T x$, where $A \in \mathbb{S}^n$ is a symmetric matrix ($A = A^T$) and $b \in \mathbb{R}^n$.

(a) Gradient $\nabla f(x)$

To find the gradient, we examine the function component-wise. The quadratic term can be written as:

$$f(x) = \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n A_{ij} x_i x_j + \sum_{i=1}^n b_i x_i$$

Differentiating with respect to x_k :

$$\frac{\partial f(x)}{\partial x_k} = \frac{1}{2} \sum_{j=1}^n A_{kj} x_j + \frac{1}{2} \sum_{i=1}^n A_{ik} x_i + b_k$$

Since A is symmetric ($A_{kj} = A_{jk}$), the first two terms are identical:

$$\frac{\partial f(x)}{\partial x_k} = \sum_{j=1}^n A_{kj} x_j + b_k = (Ax)_k + b_k$$

Collecting all partial derivatives into a vector:

$$\nabla f(x) = Ax + b$$

(b) Chain Rule with $g(h(x))$

Let $f(x) = g(h(x))$ where $g : \mathbb{R} \rightarrow \mathbb{R}$ and $h : \mathbb{R}^n \rightarrow \mathbb{R}$. By the multivariate chain rule:

$$\frac{\partial f}{\partial x_i} = \frac{dg}{dh} \frac{\partial h}{\partial x_i} = g'(h(x)) \frac{\partial h}{\partial x_i}$$

In vector form, this is the scalar $g'(h(x))$ multiplying the gradient of h :

$$\nabla f(x) = g'(h(x)) \nabla h(x)$$

(c) Hessian $\nabla^2 f(x)$

The Hessian is the matrix of second partial derivatives, or the Jacobian of the gradient. From part (a), we have $\nabla f(x) = Ax + b$. Differentiating once more:

$$\nabla^2 f(x) = \frac{\partial}{\partial x} (Ax + b) = A$$

Thus, the Hessian of a quadratic form is simply the matrix A .

(d) Hessian of $f(x) = g(a^T x)$

Let $h(x) = a^T x$. Then $\nabla h(x) = a$. Using the result from part (b):

$$\nabla f(x) = g'(a^T x)a$$

To find the Hessian, we differentiate the gradient $\nabla f(x)$ with respect to x^T :

$$\nabla^2 f(x) = \frac{\partial}{\partial x} [g'(a^T x)a] = a \frac{\partial}{\partial x} [g'(a^T x)]^T$$

By applying the chain rule to $g'(a^T x)$:

$$\nabla^2 f(x) = a[g''(a^T x)a^T] = g''(a^T x)aa^T$$

The result is a rank-1 matrix (outer product aa^T) scaled by the second derivative of g .

2. Positive Semidefinite Matrices

(a) Inner Product Property

Given a symmetric matrix $A \in \mathbb{R}^{n \times n}$, we show that $A \succeq 0$ if and only if for any $x \in \mathbb{R}^n$, $x^T Ax \geq 0$.

Proof: By the definition of positive semidefiniteness for a symmetric matrix, A is PSD if all its eigenvalues are non-negative.

- Let the spectral decomposition of A be $A = \sum_{i=1}^n \lambda_i u_i u_i^T$.
- Then, $x^T Ax = x^T (\sum_{i=1}^n \lambda_i u_i u_i^T) x = \sum_{i=1}^n \lambda_i (x^T u_i)(u_i^T x) = \sum_{i=1}^n \lambda_i (u_i^T x)^2$.
- Since $(u_i^T x)^2 \geq 0$, then $\sum \lambda_i (u_i^T x)^2 \geq 0$ for all x if and only if $\lambda_i \geq 0$ for all i .

(b) Null Space and Quadratic Form

Let $z \in \mathcal{N}(A)$ be a vector in the null space of A , such that $Az = 0$. We want to show that if A is PSD, then $z^T Az = 0$.

Solution: Substituting the null space property:

$$z^T Az = z^T (Az) = z^T (0) = 0$$

This shows that any vector in the null space results in a zero quadratic form.

(c) Positive Semidefiniteness of BAB^T

Let A be PSD and $B \in \mathbb{R}^{m \times n}$. We show that $P = BAB^T$ is also PSD.

Proof: To show $P \succeq 0$, we must verify that $y^T P y \geq 0$ for any $y \in \mathbb{R}^m$:

$$y^T (BAB^T) y = (y^T B) A (B^T y)$$

Let $x = B^T y$. Then the expression becomes:

$$x^T Ax$$

Since A is PSD, we know $x^T Ax \geq 0$ for any $x \in \mathbb{R}^n$. Thus, $y^T (BAB^T) y \geq 0$, which implies $BAB^T \succeq 0$.

3. Eigenvalues and Eigenvectors

(a) Orthogonal Matrix Properties

Let $U \in \mathbb{R}^{n \times n}$ be an orthogonal matrix, such that $U^T U = I$. We show that for any $x \in \mathbb{R}^n$, the Euclidean norm is preserved: $\|Ux\|_2 = \|x\|_2$.

Proof: Using the relationship between the L_2 norm and the inner product:

$$\|Ux\|_2^2 = (Ux)^T (Ux) = x^T U^T U x$$

Since U is orthogonal, $U^T U = I$:

$$x^T I x = x^T x = \|x\|_2^2$$

Taking the square root of both sides gives $\|Ux\|_2 = \|x\|_2$. This confirms that orthogonal transformations (like rotations) preserve the length of vectors.

(b) Eigenvectors of Symmetric Matrices

Let A be a symmetric matrix with spectral decomposition $A = U \Lambda U^T$, where $U = [u^{(1)} \dots u^{(n)}]$ is orthogonal and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$. We show that $u^{(i)}$ is an eigenvector of A .

Proof: Consider the product $Au^{(i)}$. Using the decomposition:

$$Au^{(i)} = (U \Lambda U^T) u^{(i)}$$

Since U is orthogonal, $U^T u^{(i)}$ is a vector with 1 at the i -th position and 0 elsewhere (the i -th standard basis vector e_i):

$$U^T u^{(i)} = e_i$$

Then:

$$U \Lambda e_i = U (\lambda_i e_i) = \lambda_i (U e_i) = \lambda_i u^{(i)}$$

Thus, $Au^{(i)} = \lambda_i u^{(i)}$, confirming $u^{(i)}$ is an eigenvector corresponding to eigenvalue λ_i .

(c) Eigenvalues of PSD Matrices

We show that if $A \succeq 0$ (Positive Semidefinite), then all its eigenvalues $\lambda_i(A) \geq 0$.

Proof: From part (b), we know $Au^{(i)} = \lambda_i u^{(i)}$. Using the property of PSD matrices from Question 2(a), for any x , $x^T A x \geq 0$. Let $x = u^{(i)}$ (an eigenvector):

$$(u^{(i)})^T A u^{(i)} = (u^{(i)})^T (\lambda_i u^{(i)}) = \lambda_i (u^{(i)})^T u^{(i)}$$

Since U is orthogonal, the eigenvectors are normalized: $(u^{(i)})^T u^{(i)} = \|u^{(i)}\|_2^2 = 1$. Therefore:

$$\lambda_i = (u^{(i)})^T A u^{(i)}$$

Since the quadratic form is ≥ 0 for all vectors, it must be that $\lambda_i \geq 0$ for all i .

4. Probability and Multivariate Gaussians

Let $X \in \mathbb{R}^n$ be a random variable sampled from a multivariate Gaussian distribution with mean μ and covariance $\Sigma \in \mathbb{S}_{++}^n$ (where $\Sigma \succ 0$ is positive definite). We denote this as $X \sim \mathcal{N}(\mu, \Sigma)$.

We wish to evaluate the expected value:

$$E[X^T \Sigma^{-1} X]$$

STEP 1: Use the property of the trace of a scalar Since $X^T \Sigma^{-1} X$ is a scalar (1×1 matrix), it is equal to its own trace:

$$E[X^T \Sigma^{-1} X] = E[\text{tr}(X^T \Sigma^{-1} X)]$$

STEP 2: Use the cyclic property of the trace The cyclic property states that $\text{tr}(ABC) = \text{tr}(CAB) = \text{tr}(BCA)$. Applying this here:

$$\text{tr}(X^T \Sigma^{-1} X) = \text{tr}(\Sigma^{-1} X X^T)$$

Substituting this back into the expectation:

$$E[\text{tr}(\Sigma^{-1} X X^T)] = \text{tr}(E[\Sigma^{-1} X X^T])$$

Since Σ^{-1} is a constant matrix, we can pull it out of the expectation:

$$\text{tr}(\Sigma^{-1} E[XX^T])$$

STEP 3: Relate $E[XX^T]$ to the covariance By definition, the covariance matrix Σ is:

$$\Sigma = E[(X - \mu)(X - \mu)^T] = E[XX^T] - \mu\mu^T$$

Rearranging for $E[XX^T]$:

$$E[XX^T] = \Sigma + \mu\mu^T$$

STEP 4: Substitute and simplify Substitute the expression from Step 3 into the result from Step 2:

$$\begin{aligned} \text{tr}(\Sigma^{-1}(\Sigma + \mu\mu^T)) &= \text{tr}(\Sigma^{-1}\Sigma + \Sigma^{-1}\mu\mu^T) \\ &= \text{tr}(I_n + \Sigma^{-1}\mu\mu^T) \end{aligned}$$

where I_n is the $n \times n$ identity matrix.

STEP 5: Final Evaluation Using the linearity of the trace:

$$\text{tr}(I_n) + \text{tr}(\Sigma^{-1}\mu\mu^T)$$

- $\text{tr}(I_n) = n$ (sum of ones along the diagonal).
- Applying the cyclic property again: $\text{tr}(\Sigma^{-1}\mu\mu^T) = \text{tr}(\mu^T \Sigma^{-1} \mu)$.
- Since $\mu^T \Sigma^{-1} \mu$ is a scalar, its trace is itself.

Result:

$$E[X^T \Sigma^{-1} X] = n + \mu^T \Sigma^{-1} \mu \tag{1}$$